



# MANAGEMENT PLAN SAKTENG LHAYUL SANGZEY DEYTSHEN



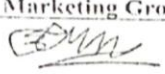
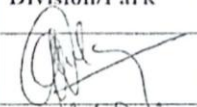
Sakteng Gewog  
Sakteng Wildlife Sanctuary (SWS)  
Trashigang  
Plan Period: 2024-2033  
(New Plan)

## Approval Sheet

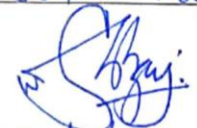
### Approval Sheet

|                                              |                                |
|----------------------------------------------|--------------------------------|
| Name of NWFP Management and Marketing Group: | Sakteng Lhayul Sangzey Detshen |
| Dzongkhag:                                   | Trashigang                     |
| Gewog:                                       | Sakteng                        |
| Village:                                     | Sakteng                        |
| No. of households:                           | 21                             |
| Area (ha):                                   | 2672.401                       |
| Plan period:                                 | January 2024-December 2034     |

### Prepared and submitted for approval:

|            | Chairperson<br>NWFP<br>Management and<br>Marketing Group                          | Gup of the<br>Gewog                                                               | Head<br>Social Forestry<br>Section of<br>Division/Park                             | Chief Forestry<br>Officer of<br>Division/Park                                       |
|------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Signature: |  |  |  |  |
| Name:      | Dorji Wangchuk                                                                    | Tshewang Hui                                                                      | Dorji Phuntsho                                                                     | Wangchuk Dorji                                                                      |
| Date:      | 19/09/2023                                                                        | 21/09/2023                                                                        | 3/10/2023                                                                          | 21-10-23                                                                            |

### Reviewed and recommended for approval:

|           | Reviewing Officer, FRPMD                                                            | Chief Forestry Officer, FRPMD                                                        |
|-----------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Name      | Kezang Wangmo                                                                       | Sondam Tobgay                                                                        |
| Signature |  |  |
| Date      | 17-11-2023                                                                          | 21/11/23                                                                             |

Approved by:



**DIRECTOR**  
Department of Forests and Park Services, MoENR



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ROYAL GOVERNMENT OF BHUTAN  
 MINISTRY OF ENERGY & NATURAL RESOURCES  
 DEPARTMENT OF FORESTS & PARK SERVICES  
 FOREST RESOURCES PLANNING AND MANAGEMENT DIVISION



DoFPS/FRPMD/FRPIS/NWFP-01/2023/ 456

21/11/2023

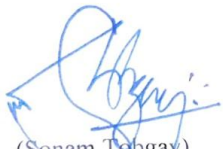
## NOTESHEET

**Subject: Approval of NWFP Management and Marketing Plan**

The Non-Wood Forest Products are an important source of livelihood for many rural communities in the country. To ensure the sustainable management of resources in the wild, management responsibilities are handed over to the communities through the formation of Non-Wood Forest Product Management and Marketing Group. Further, it is mandatory for the group to prepare a resource management and marketing plan and submit it for approval through the Chief Forestry Officer of that area. The Forest Resource Planning and Management Division had reviewed the Sakteng Lhayul Sangzey Detschen, Sakteng (new plan) Gewog under Trashigang Dzongkhag submitted by Sakteng Wildlife Sanctuary.

As per Chapter 7, Section 143 of the Forest and Nature Conservation Act of Bhutan 2023, the Department shall approve the Non-Wood Forest Product Management and Marketing Plans.

Submitted by:

  
 (Sonam Tobgay)  
 Chief Forestry Officer

Remarks:

  
 (Lobzang Dorji)  
 DIRECTOR

## **Acronyms**

|       |                                                      |
|-------|------------------------------------------------------|
| CDF   | Community Development Fund                           |
| DoFPS | Department of Forests and Park Services              |
| DT    | Dzongkhag Tshogdu                                    |
| FNCA  | Forest and Nature Conservation Act                   |
| FNCRR | Forest and Nature Conservation Rules and Regulations |
| GT    | Gewog Tshogdu                                        |
| MoENR | Ministry of Energy and Natural Resources             |
| NWFP  | Non-Wood Forest Produce                              |
| QGIS  | Quantum Geographic Information System                |
| SLSD  | Sakteng Lhayul Sangzey Deytshen                      |

## **Glossary**

|                   |                                         |
|-------------------|-----------------------------------------|
| Dzongkhag         | District                                |
| Dzongkhag Tshogdu | District Level Decision Making Body     |
| Gewog             | Sub-District                            |
| Gewog Tshogdu     | Sub-District Level Decision Making Body |

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## Part A: NWFP management and marketing plan

### 1. Background

The Sakteng Wildlife Sanctuary, established in 2003, is situated in the easternmost alpine ecosystem of Bhutan. This sanctuary encompasses the regions of Merak and Sakteng Gewog and is home to approximately 5000 semi-nomadic people. Despite the human population, the sanctuary is also a habitat for some of the rarest plant and animal species, including the Red Panda and *Meconopsis galkidyna*, the national flower of Bhutan. The sanctuary boasts a variety of medicinal and aromatic plants such as *Rhododendron setosum*, *Rhododendron anthopogon*, *Nardostachys jatamansii*, and *Juniper species*, which are commonly used in making incense sticks in the country. Unfortunately, in recent times, entrepreneurs from nearby districts have been collecting these scented plants from the sanctuary without providing any benefits to the local community.

To address this issue, the "Sustainable Production of Incense Materials at Red Panda Habitat" project was launched with the aim of benefiting both the community and incense entrepreneurs nationwide. A group was formed to produce incense raw materials at a semi-finished stage, allowing incense entrepreneurs to directly purchase these materials for making incense sticks. This arrangement benefits the local community and the incense entrepreneurs.

The people of *Merak* and *Sakteng*, known as *Brokpas*, are traditionally nomadic, relying on livestock farming for their livelihood. This group formation diversifies their sources of income and enhances their economic stability, all while ensuring that the available materials are harvested sustainably, following a well-defined management plan. This approach helps conserve resources at their source while providing maximum benefits to the local community. Further, the formation of a Non-Wood Forest Produce (NWFP) group not only strengthens and diversifies the community's income sources but also fosters collaboration and skill-sharing among the members.

In summary, the formation of this group was driven by the need to diversify the income sources for the community of Sakteng Gewog while simultaneously ensuring the sustainable harvesting of incense raw materials to preserve them for the long term.

#### 1.1. About Sakteng Lhayul Sangzey Deytshen

The name *Sakteng Lhayul Sangzey Deytshen* (SLSD) is inspired by the breathtaking beauty of the *Sakteng* villages, often compared to the concept of "*Lhyul*" or Paradise. This group comprises 21 members who voluntarily joined without any external influence. Their primary focus is the production of incense raw materials, using key ingredients like *R. setosum* (*Balu*), *R. anthopogon* (*Sulu*), and *Juniperus sp.* (*shukpu*), along with other scented plants as per the preferences of incense entrepreneurs.

In contrast to other incense groups in the sanctuary, *Sakteng Lhayul Sangzey Deytshen* exclusively deals with semi-finished raw materials for incense production and not the actual production of incense sticks. However, the group may consider venturing into incense production in the future, depending on the market demand. This expansion could add value to

the group, diversify their product range, and enhance the sustainability of their collective efforts.

## 2. Goal and objectives:

- ✓ Diversify the income source for the Highlander.
- ✓ Sustainable collection of incense raw materials from the Red Panda Habitat.
- ✓ Improve livelihood for the highlanders.

## 3. Description of NWFPs present in the area demarcated for management by the group

The area for collection or harvesting of non-wood forest produces have been identified by the group members. There are a total of 5 collection/harvesting areas named as Block I, II, III, IV, V respectively so that the group members can collect the non-wood forest produces from these identified areas.

The group members shall obtain collection permit before the commencement of the collection and the quantity of non-wood forest produces shall not exceed the quantity prescribed in this plan. The members shall also follow proper procedures and harvesting prescriptions provided in this plan.

**Table 1.** Brief description of commercial harvesting of NWFPs.

| NWFP name                              | Harvesting area (s)                                                 | Parts harvested | Use purpose (traditional & others)          | Harvesting season (months) | Estimated quantity that can be collected each season |
|----------------------------------------|---------------------------------------------------------------------|-----------------|---------------------------------------------|----------------------------|------------------------------------------------------|
| <i>Balu (R. anthopogon)</i>            | Shumling                                                            | Leaves          | Incense stick and raw dried leave or powder | September and October      | 400kg                                                |
|                                        | Jangpuensum                                                         |                 |                                             |                            | 250kg                                                |
|                                        | Jabchung                                                            |                 |                                             |                            | 500kg                                                |
|                                        | Worgho Pema Koksum                                                  |                 |                                             |                            | 300kg                                                |
|                                        | Balung                                                              |                 |                                             |                            | 300kg                                                |
| <i>Sulu (Rhododendron setosum)</i>     | Shumling                                                            | Leaves          | Incense stick and raw dried leave or powder | September and October      | 200kg                                                |
|                                        | Jangpuensum                                                         |                 |                                             |                            | 350kg                                                |
|                                        | Jabchung                                                            |                 |                                             |                            | 400kg                                                |
|                                        | Worgho-Pema Koksum                                                  |                 |                                             |                            | 200kg                                                |
|                                        | Balung                                                              |                 |                                             |                            | 450kg                                                |
| <i>Baa-po (Saussurea gossipiphora)</i> | Shumling<br>Jangpuensum<br>Jabchung<br>Worgho-Pema Koksum<br>Balung | Flower          | Incense                                     | August and September       | 5kg                                                  |
| <i>Gya-po</i>                          | Shumling<br>Jangpuensum                                             | Leave and       | Incense                                     | August and                 |                                                      |

|                                         |                                                                                                                                                                                                                                   |                   |         |                                          |                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------|------------------------------------------|-------------------------|
| <i>(Tanacetum atkinsonii)</i>           | Jabchung<br>Worgho- Pema<br>Koksum Balung                                                                                                                                                                                         | stem              |         | September                                | 3kg                     |
| <i>Yulu (Cassiope fastigiata)</i>       | Shumling<br>Jangpuensum<br>Jabchung Worgho-<br>Pema Koksum<br>Balung                                                                                                                                                              | Leave and<br>stem | Incense | August and<br>September                  | 2kg                     |
| <i>Atrung karmo</i>                     | Shumling<br>Jangpuensum<br>Jabchung Worgho-<br>Pema Koksum<br>Balung                                                                                                                                                              | Leave and<br>stem | Incense | August and<br>September                  | 2kg                     |
| <i>Shalley karmu (Ajania nubigenum)</i> | Shumling<br>Jangpuensum<br>Jabchung Worgho-<br>Pema Koksum<br>Balung                                                                                                                                                              | Whole<br>body     | Incense | September<br>and<br>October              | 2kg                     |
| <i>Pangpoe (Nardostachys jatamansi)</i> | Tsholung<br>Gonpa Lamai<br>Traptru                                                                                                                                                                                                | Root              | Incense | October<br>and<br>November               | 2kg                     |
| <i>Ringu (Swertia wardii)</i>           | Shumling<br>Jangpuensum<br>Jabchung                                                                                                                                                                                               | Root              | Incense | October<br>and<br>November               | 1kg                     |
| <i>Shub Shing (Juniperus species)</i>   | Phugshenang,<br>Donglok,<br>Tholongshing,<br>Brogsar, Kenzer,<br>Manipem,<br>Shengue,<br>Marzim, Balung,<br>Salay, Chamgen,<br>Lhabrateng,<br>Dakpashisa,<br>Tshumdepang,<br>Domshisa,<br>Renakpu,<br>Mesakteng and<br>Nakarchung | Leave/<br>Needles | Incense | Based on<br>availability<br>in the field | 1 Truckload<br>per year |

#### 4. Harvesting prescriptions for sustainable NWFP harvesting

The harvesting prescriptions and guidelines are referred from the “Framework for management and marketing of Non-Wood Forest Produce and Forest and Nature Conservation Code of Best Management Practices of Bhutan. The guidelines for responsibly harvesting Non-Wood Forest Products (NWFPs) are as follows:

#### 4.1. Balu (*Rhododendron anthopogon*)

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                     | <i>R. anthopogon</i> , also known as the dwarf rhododendron, is a flowering plant classified within the Ericaceae family. Typically, it is located on inclines, rocky outcrops, and within dense vegetation. This plant is indigenous to Pakistan and can be located in the Himalayan areas. It is valued for its utilization in the creation of essential oils with fragrant scents and for its role in the production of incense. |
| Harvesting Guidelines and Harvesting Techniques | Harvest a maximum of 20% of the leaves and twigs from a single plant. Employ sharp knives or scissors to trim the branches when collecting leaves from the individual plant.                                                                                                                                                                                                                                                        |
| Harvesting Limit                                | The collection of <i>R. anthopogon</i> must be as prescribed in this management plan.                                                                                                                                                                                                                                                                                                                                               |

#### 4.2. Sulu (*Rhododendron setosum*)

|                                                 |                                                                                                                                                                                                                                  |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                     | <i>R. setosum</i> is a compact flowering shrub classified within the Ericaceae family. Its blooming period typically commences between May and July. <i>R. setosum</i> is alternatively referred to as the bristly Rhododendron. |
| Harvesting Guidelines and Harvesting Techniques | Gather a maximum of 20% of the leaves and twigs from a single plant. Utilize sharp knives or scissors to trim the branches when harvesting leaves from the individual plant.                                                     |
| Harvesting Limit                                | The collection of <i>R. setosum</i> must be as prescribed in this management plan.                                                                                                                                               |

#### 4.3. Baa-po (*Saussurea gossipiphora*)

|                                                 |                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                     | This plant, often referred to as the snowball plant, is a perennial herbaceous plant recognized for its medicinal attributes. It originates from the Himalayan region and thrives at elevations ranging from 4300 to 5600 meters. These plants serve as source materials in the production of incense in factories. |
| Harvesting Guidelines and Harvesting Techniques | As the entire stem, including the leaves, is employed in incense production, the stem will be removed from ground level using a knife, sizable scissors, and pruning shears.                                                                                                                                        |
| Harvesting Limits                               | The collection of all these herbaceous plants must be as prescribed in this management plan.                                                                                                                                                                                                                        |

#### 4.4. Gya-po (*Tanacetum atkinsonii*)

|                                                 |                                                                                                                                                                                                          |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                     | These plants are categorized within the Asteraceae family, a group of flowering plants, and are commonly referred to as tansies. They primarily consist of perennial herbs and emit a distinctive aroma. |
| Harvesting Guidelines and Harvesting Techniques | To produce incense, the entire stem and leaves are utilized. The stem is typically removed at ground level using tools such as a knife, large scissors, or pruning shears.                               |
| Harvesting Limits                               | The collection of all these herbaceous plants must be as prescribed in this management plan.                                                                                                             |

#### 4.5. Yulu (*Cassiope fastigiata*)

|                                                 |                                                                                                                                                                                      |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                     | This diminutive shrub, a member of the Ericaceae family, thrives as an undergrowth shrub in high-elevation forests. It can be located at altitudes ranging from 3000 to 4000 meters. |
| Harvesting Guidelines and Harvesting Techniques | To create incense, the entire stem, along with its leaves, is harvested by cutting it down from ground level using implements such as a knife, large scissors, or pruning shears.    |
| Harvesting Limits                               | The collection of all these herbaceous plants must be as prescribed in this management plan.                                                                                         |

#### 4.6. Atrung karmo (*Ajania nubigenum*)

|                                                 |                                                                                                                                                                              |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                     | This plant is a member of the Asteraceae family, commonly referred to as the Bhutanese daisy. It is typically found at altitudes ranging from 3600 to 4800 meters.           |
| Harvesting Guidelines and Harvesting Techniques | To produce incense, the entire stem, along with its leaves, is harvested by cutting it down at ground level using tools such as a knife, large scissors, and pruning shears. |
| Harvesting Limits                               | The collection of all these herbaceous plants must be as prescribed in this management plan.                                                                                 |

#### 4.7. Shalley karmo (*Thamnolia vermicularis*)

|                                                 |                                                                                                                                                                |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                     | <i>Thamnolia</i> is a lichen genus found in the Icmadophilaceae family, commonly known as "white worms".                                                       |
| Harvesting Guidelines and Harvesting Techniques | The entire lichen body is employed for making incense, and it is recommended to gather them by hand. The use of any equipment is not allowed for this purpose. |
| Harvesting Limits                               | Only 20% of the total resources available will be collected from the area                                                                                      |

#### 4.8. Ringu (*Swertia wardii*)

|                                                 |                                                                                                                                                                                                                        |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                     | This perennial plant typically stands at a height of 30-80 cm and features blackish, short rhizomes with a few slightly fleshy rootlets. It is commonly located within an altitudinal range of 3800-5200 meters.       |
| Harvesting Guidelines and Harvesting Techniques | Gather the plants only after the seed dispersal process has begun. Avoid collecting the plants late in the season, particularly in February, as this could pose a significant risk to the germination of new seedlings |
| Harvesting Limits                               | Only 20% of the total resources available will be collected from the area.                                                                                                                                             |

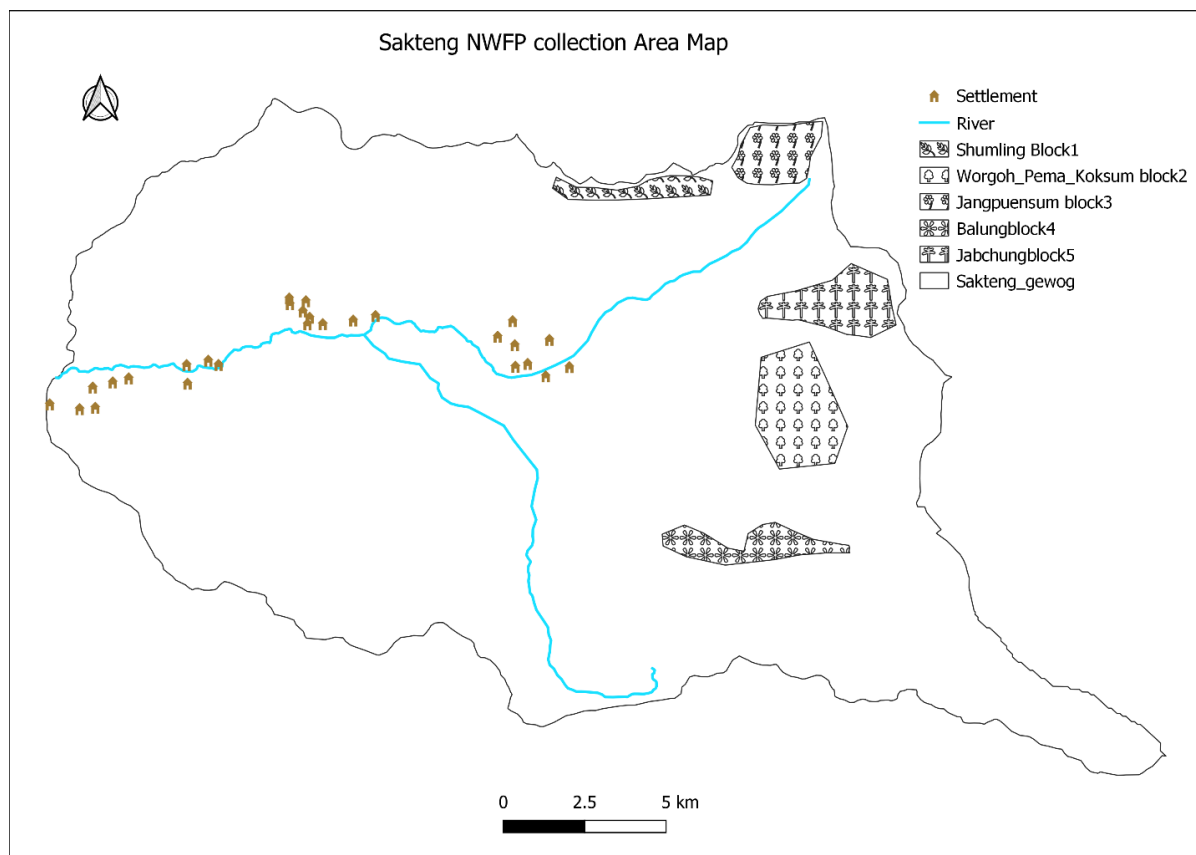
#### 4.9. Pangpo (*Nardostachys jatamansi*)

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                     | <i>Nardostachys jatamansi</i> is an upright, hairy, perennial herb, standing at a height of 10–60 cm, characterized by a robust, woody main root. The rootstock is thick, elongated, and covered with the remnants of withered leaves' petioles. The stems are typically hairy above and become less hairy below. This herb thrives at elevations ranging from 3000 to 5000 meters. |
| Harvesting Guidelines and Harvesting Techniques | To harvest the rhizome, carefully dig it up using a sharp knife. Make sure to leave the mother plant in place with a small section of the rhizome remaining in the soil, as this part will regenerate in the following year. Avoid uprooting the entire plant.                                                                                                                      |
| Harvesting Limits                               | The collection of <i>N. Jatamansii</i> must be as prescribed in this management plan.                                                                                                                                                                                                                                                                                               |

#### 4.10. Leaves or Needles of *Juniperus species* (Shupshing)

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                          | These plants can be either trees, reaching heights of up to 20 meters, or shrubs, which can reach up to 30 meters in height. They have needle-like leaves that are arranged along the main shoot and 4-ranked scale-like leaves on the branches. These plants have flaky brown bark, and their scales overlap, producing slender leafy shoots. The female cones typically consist of 3-8 scales that become fleshy and cohere to form berry-like fruits. When young, these fruits are globose, and they become ovoid when ripe. The seeds are usually solitary, thick, and hard. |
| Harvesting Guidelines and Techniques | The needle can be harvested based on the availability in the growing areas without hampering the growth of trees. For collection of needles, trees should not be harvested.                                                                                                                                                                                                                                                                                                                                                                                                      |
| Harvesting Limits                    | The collection of dried and fallen <i>Juniper</i> should be limited to areas along the transmission lines and the Gewog center road. However, depending on availability, collection may also be conducted in other specified areas designated for collection.                                                                                                                                                                                                                                                                                                                    |

## 5. Google/QGIS map of the NWFP harvesting areas



**Figure 1.** Map of NWFP collection area

## 6. Marketing and sales plan

The group aims to distribute their products to the nearest incense stick factory, local shops, and visitors. The detailed marketing and sales plans are provided as attachments below.

**Table 2.** Marketing and sales plan

| NWFP                                  | Potential market                                                         | Estimated quantity for sale/ season (unit) | Form of Products (dry/fresh) | Expected price/ unit | Total (Nu) |
|---------------------------------------|--------------------------------------------------------------------------|--------------------------------------------|------------------------------|----------------------|------------|
| <i>Balu (Rhododendron anthopogon)</i> | NWFP traders, incense factories, local people, shops, and local dealers. | 600kg                                      | Dry                          | 80                   | 48000      |
| <i>Sulu (Rhododendron setosum)</i>    |                                                                          |                                            |                              |                      |            |
| <i>Juniper species</i>                | NWFP traders, incense factories                                          | 2000kg                                     | Dry                          | 40                   | 80000      |

|                                                            |                                                                          |          |              |    |         |
|------------------------------------------------------------|--------------------------------------------------------------------------|----------|--------------|----|---------|
| <i>Balu (Rhododendron anthopogon)</i>                      | NWFP traders, incense factories, local people, shops, and local dealers. | 6000 pkt | Dry (Powder) | 40 | 240,000 |
| <i>Sulu (Rhododendron Setosum)</i>                         |                                                                          |          |              |    |         |
| <i>Baa-po (Saussurea gossipiphora)</i>                     |                                                                          |          |              |    |         |
| <i>Gya-po (Tanacetum atkinsonii)</i>                       |                                                                          |          |              |    |         |
| <i>Yulu (Cassiope fastigiata)</i>                          |                                                                          |          |              |    |         |
| <i>Atrung karmo (Ajania nubigenum)</i>                     |                                                                          |          |              |    |         |
| <i>Shalley karmo (Thamnia vermicularis)</i>                |                                                                          |          |              |    |         |
| <i>Ringu (Swertia wardii)</i>                              |                                                                          |          |              |    |         |
| <i>Pangpo (Nardostachys jatamansi)</i>                     |                                                                          |          |              |    |         |
| <i>Leaves and needles of Juniperus species (Shupshing)</i> |                                                                          |          |              |    |         |

## 7. Marketing arrangements

The group will individually collect raw materials from their respective sources and sell them to the group for incense material processing. A reasonable price for the raw materials collected will be paid to the members from the group's revenue. When instructed by the executive members of the group, all members must be present for processing the raw materials and will receive Nu. 60 (Sixty) per hour for their labour. If the labour work takes an entire day, a minimum of Nu. 500 (Five hundred) will be paid to the members who are present for the processing works.

The executive committee of the group will be responsible for finding a market and selling the product. They will be compensated with Nu. 4000 for the chairman and Nu. 3000 each for the Secretary and Treasurer annually as an incentive for selling the product on the group's behalf. The executive committee will decide if all members need to be gathered for work that benefits the group without any labour charges.

**Table 3.** Procedures for NWFP harvesting, marketing and sharing of income

| Activity | Procedures                                                                                                                                                                                                                       |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Once the chairperson of the group obtains the necessary permit from the nearest forest office, all group members are accountable for collecting the raw materials from the forest in the quantities specified in this management |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Harvesting of products                         | <p>plan. The chairperson will determine the specific quantity for each member to collect based on this plan. In cases where members are unwilling to collect, the chairperson may delegate non-members to collect on their behalf and expenses will be covered from the group's revenue.</p> <p>Payment for group members involved in raw material collection will be disbursed at the end of the year during the closing of accounts. The chairperson is responsible for maintaining a detailed stock register for the materials collected.</p>                                                                                                                                                                      |
| Postharvest practices                          | The incense materials will be collected individually by the group members in accordance with the specified management plan from the designated areas outlined in this plan. Subsequently, these materials will be processed at the incense raw material production farmhouse.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Marketing of products                          | The Chairperson and Secretary of the group will assume the responsibility of marketing the products produced by the group. They will actively seek out and explore all available markets for selling the group's products.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Income sharing                                 | <p>Every group member will receive compensation for collecting raw materials and working on the production of semi-finished products. However, these payments will be distributed at the end of the year, based on the revenue generated by the group.</p> <p>After covering the costs of raw material collection, processing, and marketing, any remaining revenue generated will be deposited into a joint account. At the end of the five years, 50% of the profits will be divided equally among the active group members, while the remaining 50% will be retained in the joint account to ensure the sustainability of the group. Inactive members will not be eligible for a share of the group's profits.</p> |
| Raising of community development fund          | The chairperson, in the presence of all members, will make the decision regarding whether the group can allocate a portion of its revenue to contribute to community development funds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Payment to committee members for their service | Each year, the Chairperson will receive Nu. 4000, while the Secretary and Treasurer will each receive Nu. 3000 from the group's generated revenue as an incentive for their role in marketing the products.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## 8. Benefits to the members

Annually, the group is expected to generate revenue of Nu. 368,000 (Three Lakh Sixty-Eight Thousand) only. After accounting for expenses of approximately Nu. 34,000 (Thirty-Four Thousand) only for packaging and marketing the product, the group anticipates a net group income of about Nu. 334,000 (Four Lakh Six Thousand) only.

**Table 4.** Financial projection of the benefits to the members from NWFP sales.

| Sl. No   | Particulars                                                                              | Products (names)                                  |                   |                              | Total Amount (Nu) |
|----------|------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------|------------------------------|-------------------|
|          |                                                                                          | Semi-finished incense raw materials (powder form) | Balu & Sulu (dry) | Dried Juniper leaves /trunks |                   |
| <b>1</b> | <b>Revenues</b>                                                                          |                                                   |                   |                              |                   |
| a        | Total quantity (unit)                                                                    | 6000pkt                                           | 600kg             | 2000kg                       | -                 |
| b        | Market price (Nu per unit)                                                               | 40                                                | 80                | 40                           | -                 |
| <b>c</b> | <b>Total revenues from forest product sales (a*b)</b>                                    | <b>240,000</b>                                    | <b>48,000</b>     | <b>80,000</b>                | <b>368,000</b>    |
| <b>A</b> | <b>Total revenues</b>                                                                    | <b>240,000</b>                                    | <b>48,000</b>     | <b>80,000</b>                | <b>368,000</b>    |
| <b>2</b> | <b>Expenses</b>                                                                          |                                                   |                   |                              |                   |
| a        | Packaging materials                                                                      | 10,000                                            | 3000              | 4000                         | 17000             |
| b        | Royalty                                                                                  | 2000                                              |                   |                              | 2000              |
| c        | Transportation costs                                                                     | 5000                                              | 5000              | 5000                         | 15000             |
| d        | Compensation fee to the executive committee for marketing products                       | NA                                                | NA                | NA                           |                   |
| e        | Contribution to the community development fund                                           | NA                                                | NA                | NA                           |                   |
| f        | Incidental expenses                                                                      | NA                                                | NA                | NA                           |                   |
| <b>B</b> | <b>Total expenses</b>                                                                    | <b>17000</b>                                      | <b>8000</b>       | <b>9000</b>                  | <b>34,000</b>     |
| <b>C</b> | <b>Total benefit (net income) of group members (total revenues A – total expenses B)</b> | <b>223,000</b>                                    | <b>40,000</b>     | <b>71,000</b>                | <b>334,000</b>    |
| <b>D</b> | <b>Average income to group members (C/ no of group members)</b>                          |                                                   |                   |                              | <b>15905</b>      |
| <b>E</b> | <b>Price per unit paid to group members (C/1a)</b>                                       | <b>37.17</b>                                      | <b>66.67</b>      | <b>35.5</b>                  | <b>139.34</b>     |

## 9. Benefits to the group

The revenue generated will be deposited into the group's joint account, and withdrawals can only be made with the signatures of three executive members of the group.

Group members will have the option to avail themselves of a loan facility of up to Nu. 20,000 during financial difficulties, with an interest rate of 5%. The loan amount may be adjusted based on the financial status of the group. Subject to budget availability, the Chairman, in the presence of all members, may decide to provide financial support in the event of an unfortunate incident involving a member.

The Chairman is responsible for maintaining the meeting agenda, which will be signed by the members to provide justification for decisions made during meetings.

**Table 5.** Financial projection of the benefits to the group, the community development fund.

| Sl. No    | Particulars                                                                    | Amount (Nu)    |
|-----------|--------------------------------------------------------------------------------|----------------|
|           | <b>Revenues</b>                                                                |                |
| 1         | Membership fees                                                                |                |
|           | a. Initiation of new group (collected at 1 <sup>st</sup> year)- 500*21         | 10,500         |
|           | b. Annual Contribution (2 <sup>nd</sup> year onwards)- 250*21                  | 5250           |
| 2         | Contribution to the Community Development Fund (CDF) from sales of product 1   | 223,000        |
| 3         | Contribution to the CDF from sales of product 2                                | 40,000         |
| 4         | Contribution to the CDF from sales of product 3                                | 71,000         |
| <b>A1</b> | <b>Total revenues excluding annual contribution from 2<sup>nd</sup> year</b>   | <b>344,500</b> |
| <b>A2</b> | <b>Total revenues excluding membership fee for initiation of the new group</b> | <b>339,250</b> |
|           | Expenses (only include expenses that will be paid from the group funds)        | 167,000        |
| <b>B</b>  | <b>Total Expenses</b>                                                          | <b>167,000</b> |
| <b>C1</b> | <b>Balance: Total revenues (A1)– total expenses</b>                            | <b>177,500</b> |
| <b>C2</b> | <b>Balance: Total revenues (A2)– total expenses</b>                            | <b>172,250</b> |

## **Part B: Group's BY-LAW**

### **10. Roles and responsibilities of NWFP management & marketing group members**

#### **10.1. Rights:**

1. All properties, including machinery and permanent structures, are owned by the members of SLSD.
2. Any product produced by the unit belongs to the group.
3. All members have the right to receive funds or amounts generated from the sale of products.
4. SLSD members have the right to collect resources from the areas specified in the management plan.
5. All members have the right to inquire about the funds generated from the sales of products from this unit.
6. All members have the right to provide constructive feedback for the improvement of the group.
7. All active members are entitled to a share from the group at the end of the plan period

#### **10.2. Responsibilities:**

1. Every member shall be responsible for taking care of the properties registered under this unit.
2. All members shall be accountable for the loss of accessories.

3. All members shall be responsible for exploring market for the product and inform the chairperson accordingly.
4. All members should engage in the collection of raw materials as per the prescribed methodology in this management plan.
5. Every member shall be responsible for managing and collecting the resources sustainably from the designated management area.
6. All members shall abide by this by-law.

## **11. Roles and responsibilities of NWFP management and marketing group committee**

### **11.1. Chairman**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b><i>Name of Chairman &amp; contact no.</i></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b><i>Dorji Wangdi (17954752)</i></b> |
| <b>Responsibilities of Chairperson:</b> <ul style="list-style-type: none"> <li>❖ Organize committee meetings or consultations.</li> <li>❖ Lead in the formulation of annual work plans and daily activities.</li> <li>❖ Guest entertainment and facilitation.</li> <li>❖ Explore market opportunities for the sale of products.</li> <li>❖ Co-ordinate and build interpersonal relationships with the stakeholders.</li> <li>❖ Co-ordinate in the unit improvement and developmental activities.</li> <li>❖ Financial management and verifications.</li> <li>❖ Resolve any conflict among members.</li> <li>❖ Maintain daily/ hourly registers of the members who work for the group in processing the raw materials.</li> <li>❖ Represent the group in meetings and workshops conducted by relevant stakeholders.</li> <li>❖ Oversee the group members going against these By-laws and management plans.</li> <li>❖ Entertain members to share their grievances and give a critical judgment.</li> </ul> |                                       |

### **11.2. Secretary**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b><i>Name of Secretary and Contact No.</i></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b><i>Leki Drakpa (17514761)</i></b> |
| <b>Responsibilities of Secretary:</b> <ul style="list-style-type: none"> <li>❖ Assist or lead any committee meetings or consultations.</li> <li>❖ Assist with annual work plans and daily activities.</li> <li>❖ Guest entertainment and facilitation.</li> <li>❖ Assist in exploring market opportunities.</li> <li>❖ Assist in coordinating and building interpersonal relationships with the stakeholders.</li> <li>❖ Assist to co-ordinate unit improvement and development activities.</li> <li>❖ Assist in financial management and verifications.</li> <li>❖ Assist to resolve any conflict among members.</li> <li>❖ In the absence of the Chairperson due to unavoidable circumstances Secretary will take up the role of Chairperson.</li> <li>❖ Record keeping and office management.</li> </ul> |                                      |

### 11.3. Treasurer

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <i>Name of treasurer and contact No.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>Rinchen Dema (17421921)</i> |
| Responsibilities of Treasurer: <ul style="list-style-type: none"><li>❖ Maintain records of yearly financial transactions.</li><li>❖ Receiving and issuance of money receipt.</li><li>❖ Collection of membership fees and fines.</li><li>❖ Take ownership of cashbooks and bank balance books.</li><li>❖ Deposit group funds and declare financial status to the committee members during meetings every six months.</li><li>❖ Assist chairman and secretary as and when needed</li></ul> |                                |

### 12. Tenure and incentives to committee members

|                                     |                                                                                                                                                                                                                                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tenure of committee member          | The committee members shall be changed every 3 years.<br>The committee members shall be elected by the group members.                                                                                                                                             |
| Handing taking charge               | During the change of the committee members, a proper handing taking shall be in place and signed by the previous and newly elected committee members. This shall support the group members and committee members in updating and auditing the group's activities. |
| Incentives to the committee members | Chairperson- Nu.4000 on annual basis<br>Treasurer and Secretary- Nu.3000 each on annual basis.<br>The incentives shall be paid from the revenue generated through sales of the products.                                                                          |

### 13. Sources for generation of group fund

|                                    |                                                                                                                                                         |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initiation of group                | For the initiation of the group, members shall contribute Nu. 500 as the membership fee to the group                                                    |
| Annual contribution                | A sum of Nu. 250 will be contributed annually as a membership contribution to the group.                                                                |
| Registration of new members        | The new members will be entertained upon paying Nu. 5000 (Five thousand) as membership fee.                                                             |
| Discontinue of annual contribution | The chairman in the presence of the members would decide to discontinue the annual membership fee if the revenue generated by the group is outstanding. |

## 14. Fund Management and utilization

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group expenditure | <ul style="list-style-type: none"> <li>✓ Expenditure for harvesting, transportation, and production costs, including wages for the processing of raw materials and collection of the viable incense plants from the sources to the processing unit and then to the nearest market.</li> <li>✓ Purchase of office stationery/properties.</li> <li>✓ Maintenance of production unit, the group shall improve the enterprise infrastructure and technology as required and decided by the members.</li> <li>✓ For charity purposes/donation</li> </ul>                                                    |
| Loan facilities   | <ul style="list-style-type: none"> <li>✓ The group will allow a small grant loan amounting to Nu. 20,000 (Twenty thousand) only to interested stakeholders.</li> <li>✓ The repayment of the loan should be done within one year from the date of loan sanctioned.</li> <li>✓ The interest rate of 5% per month shall be collected from the borrower.</li> <li>✓ The chairman in the presence of all members will decide the loan amount as per the financial status of the group.</li> <li>✓ Annually, the chairperson shall conduct an income generation meeting cum benefit disbursement.</li> </ul> |

## 15. Conflict resolution

|                             |                                                                                                                                                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conflict among communities  | In the event of disputes or conflicts among communities regarding access to specific areas for harvesting of non-wood forest products (NWFP), the matter will be escalated to the <i>Gewog Tshogdu</i> (GT) and <i>Dzongkhag Tshogdue</i> (DT) for a final decision and resolution. |
| Conflict within NWFP groups | Any conflicts or disputes within the NWFP Management Group that cannot be resolved by the executive members of the group will be escalated and forwarded to the <i>Gewog</i> Administration for resolution.                                                                         |

## 16. Meeting and purpose

|                   |                                                                                                                                                                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Committee meeting | The committee will hold quarterly meetings to assess the marketing strategy and address other relevant group matters. Decisions made during these committee meetings should be communicated to the remaining members through any social communication medium used by the group. |
| General meeting   | The Chairperson should conduct a General meeting annually with the group members to report on the fund balance and group activities. During these meetings, the Chairperson is responsible for disclosing the expenses and the revenue generated by the group for the year.     |

## 17. Penalties

|                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-payment of annual fees                                                                    | The chairperson will issue two written warnings to the individual regarding their outstanding fees. If the individual fails to pay the fees even after receiving reminders from the chairperson, he/she will be required to exit the group by the deadline specified by the chairperson.                                                                                                                      |
| Failure to attend the meeting/<br>mass unpaid work or coming to<br>the meeting very late      | A late fee of Nu.50/- and an absentee fee of Nu.500. However the payment of fines and late fees would be exempted in unavoidable circumstances.                                                                                                                                                                                                                                                               |
| Collection and sale of NWFPs<br>in non-adherence to group-<br>agreed norms                    | 1 <sup>st</sup> Offence fine: Nu.500/-,<br>2 <sup>nd</sup> Offence fine: Nu.1000/-, and<br>3 <sup>rd</sup> offense and above: SLSD members to come up with a<br>resolution.<br>Confiscated product shall be sold at prevailing market rate.                                                                                                                                                                   |
| Embezzlement of group funds<br>by committee member                                            | 5 percent penalty for the misused amount in addition to the<br>principal amount and he/she must resign from the committee<br>member.                                                                                                                                                                                                                                                                          |
| Exiting the group                                                                             | The Chairman in the presence of all the members may allow<br>group members to leave the group on payment of Nu. 2500 (Two<br>thousand five hundred) only as group contribution. The member<br>may be allowed to leave without payment of group contribution<br>if the member leaves the group due to unavoidable<br>circumstances.                                                                            |
| Personal Use of Group<br>Properties                                                           | 1 <sup>st</sup> Offence: Fine of Nu. 5000 (five thousand)<br>2 <sup>nd</sup> Offence: Fine of Nu. 10,000<br>3 <sup>rd</sup> Offence: Fine of Nu. 20,000 and compulsory retirement from<br>the group.<br>If the offender fails to pay the fines as per this by-law, he or she<br>shall be dealt with existing rules and regulations of Bhutan.<br>Confiscated product shall be sold at prevailing market rate. |
| If the members fail to comply<br>with the bylaws of the<br>management plan                    | 1 <sup>st</sup> Offence fine: Nu.500/-,<br>2 <sup>nd</sup> Offence fine: Nu.1000/-, and<br>3 <sup>rd</sup> offense and above: SLSD members to come up with a<br>resolution.                                                                                                                                                                                                                                   |
| Disobeying/failure to execute<br>the work assigned to the<br>members by executive<br>members. | Non-adherence to the work assigned by the executive members<br>for the interest of the group, Nu. 500 shall be paid by the person<br>to the group.                                                                                                                                                                                                                                                            |

## 18. Power of the Department

- The Department of Forests and Park Services shall inspect, monitor, review the records and supervise the activities of the group as per the Forest and Nature Conservation Act of Bhutan 2023 and its rules and regulations.
- The DoFPS has the right to terminate, cancel or reinstate the group in case of any violation of the management plan prescription and in the event of inability to carry out the activities as per the plan.

**Table 6.** Lists of NWFP group members.

| Sl. No | Name of the household representative | CID No      | House No | Thram No |
|--------|--------------------------------------|-------------|----------|----------|
| 1      | Dorji Wangdi                         | 11303002609 | 348      | 139      |
| 2      | Pema Khandu                          | 11511001030 | 389      | 325      |
| 3      | Leki Drakpa                          | 11511001250 | 407      | 452      |
| 4      | Sangay Dorji                         | 11511001380 | 96       | 68       |
| 5      | Dorji Pempa                          | 11511000832 | 174      | 433      |
| 6      | Pema Tashi                           | 11511001783 | 380      | 931      |
| 7      | Lobzang Norbu                        | 1150700560  | 100      | 47       |
| 8      | Rinchen Tshering                     | 11511002477 | 99       | 161      |
| 9      | Sangay Tenzin                        | 11511000742 | 123      | 149      |
| 10     | Phurpa Dema                          | 11511000947 | 68       | 206      |
| 11     | Dema Yangzom                         | 11511001554 | 246      | 884      |
| 12     | Jamyang Dema                         | 11511001825 | 203      | 460      |
| 13     | Tshering Choden                      | 11511002445 | 328      | 458      |
| 14     | Rinchen Dema                         | 11511000803 | 358      | 320      |
| 15     | Dechen Choden                        | 11511001829 | 204      | 105      |
| 16     | Pema Chozom                          | 11511001171 | 70       | 260      |
| 17     | Pema Yangchen                        | 11511001681 | 100      | 137      |
| 18     | Nima Dema                            | 11511000827 | 134      | 57       |
| 19     | Karma Yangzom                        | 11511000963 | 371      | 925      |
| 20     | Phurpa Dema                          | 11511001162 | 68       | 39       |